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The Editor Ocean and Coastal Research

Dear Editor,

I am submitting my manuscript, "DOUBLE JEOPARDY: The Vulnerability Spiral of Compound Climate Risk Across 5 Islands" (Running Title: Compound Climate Vulnerability Across Small Islands), for consideration as an Original Article in Ocean and Coastal Research.

This study tests, across 5 islands in 3 ocean basins (Maldives, Lakshadweep, Seychelles, Fiji, and the Canary Islands), whether the coastal ecosystems that provide protective buffering against sea level rise are themselves degrading uniformly. Treating mangrove decline and coral reef degradation as 2 separate, independently tested hypotheses rather than a single combined category, I find that coral thermal stress is rising across most of the sample while mangrove extent shows no measurable decline, so ecosystem degradation does not happen uniformly by type. Combining physical exposure with the change in coral thermal stress between early and recent reference periods into a composite Compound Vulnerability Score, and separately testing whether protected area coverage tracks that empirically verified risk, produces results with direct relevance for how climate adaptation and conservation resources are prioritized across islands and ecosystem types.

This manuscript is original, has not been published previously, and is not under consideration for publication in any other journal. I am the sole author and have approved this submission.

In line with the journal's policy on generative artificial intelligence, I disclose that Claude (Anthropic) was used during the preparation of this manuscript in 2 ways: for grammar, punctuation, and prose level review, including identifying internal inconsistencies and overstated claims relative to the underlying methodology; and as a coding assistant during the analysis, to help identify bugs in the data processing pipeline. All AI-assisted suggestions were reviewed by me, and I take full responsibility for the accuracy and interpretation of the manuscript's content, data, findings, and citations. The full AI Use Disclosure appears in the manuscript itself.

I declare no conflicts of interest related to this work. All analysis scripts and derived outputs supporting this study are publicly available in the [GitHub repository](#) and archived on [Zenodo](#); source datasets remain publicly available from their original providers. A preprint is posted on [EarthArXiv](#).

Thank you for your time and consideration. I look forward to your response.

Sincerely,  
Sakshi D. Maske